

Pradip Sahane

Senior Manager – R&D (Medical Robotics & Industrial Automation)

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📍 At Sirasagaon mandap, Jalna, Maharashtra, India, Pincode: 431135



Career Summary

Proud core R&D contributor to the Misso Orthopedic Robotic System — India's first compact intelligent orthopedic surgical robot — with 1000+ successful live cases. Senior R&D leader with 8+ years of experience in robotics, AI/ML, industrial automation, and medical device compliance (ISO 13485, IEC 62304, IEC 60601, ISO 14971, IEC 62366, CE MDR). At Meril Healthcare, spearheaded product maturity, clinical safety, and adoption of surgical robotics systems. Previously at KUKA India, implemented vision-guided robotic solutions and ML applications for industrial automation clients including TATA, Mahindra, and Volkswagen. At Grindmaster Machines, developed innovative automation solutions including a low-cost COVID ventilator deployed in hospitals. Expert in system integration, motion planning, surgeon training, clinical validation, CAPEX/resource planning, Jira automation, and delivery of parallel orthopedic projects. Proven track record in managing multidisciplinary teams, vendor negotiations, and the complete product lifecycle from R&D to post-market surveillance.

Technical Skills

- **AI & Data:** Machine Learning, Deep Learning (PyTorch, TensorFlow), Data Pipelines, Model Evaluation, Active Learning
- **Robotics & Motion:** Robot Motion Planning, Kinematics, RoboDK, RobotMaster, ROS, RobotStudio
- **Industrial Automation:** ABB, KUKA, FANUC, UR, Dobot; Mitsubishi/Siemens/Allen Bradley/Beckhoff
- **Vision & Perception:** Cognex A5000, 3D vision, OpenCV, YOLO, sensor integration
- **Medical Device & QMS:** ISO 13485, IEC 62304, IEC 60601, ISO 14971, IEC 62366, CE MDR, FDA knowledge
- **Software & Tools:** C#, WPF, Python, MATLAB/Octave, Mastercam, Git, Jira, Visual Studio

Experience

Meril Healthcare — *Senior Manager, R&D*

22 Jun 2023 – 25 Jul 2025

- Core R&D contributor to the Misso Orthopedic Robotic System; responsible for product maturity, clinical safety, and adoption.
- Supported and validated 1000+ live surgical cases; led clinical escalation and post-market surveillance inputs.
- Headed pan-India technical & clinical support: surgeon training, proctor scheduling, in-OR troubleshooting.
- Managed multidisciplinary teams; responsible for team building, resource hiring and mentoring.
- Ownership of CAPEX/OPEX budgeting, vendor negotiation and timeline delivery.
- Implemented Jira automation, Gantt planning, project libraries and resource allocation matrices for 3 live orthopedic R&D projects.
- Directed development of C#/WPF surgical planning software, robot motion planning and calibration flows; enforced traceability and change control.
- Authored V&V plans, test scripts, clinical validation protocols, and regulatory submissions (CE/FDA).

Meril Healthcare — *Manager, R&D*

Apr 2023 – Jul 2025

- Led early R&D iterations, simulation & validation of surgical flows and clinical safety features.
- Implemented system testing scripts, lab validation, and improved CR-surgery workflows & safety interlocks.

- Coordinated cross-functional pilots and pre-clinical cadaver trials; documented results in DHF.

KUKA India — *Robotic Vision Application Engineer (Customer Service)*

Apr 2021 – Jul 2021

- ROS development, vision system integration and ML implementation for industrial automation.
- Designed and commissioned vision-guided robotic cells for material handling, spot welding, grinding and deburring.
- Implemented ML/computer-vision for part detection, seam tracking and path optimization.
- Delivered installations, system upgrades and on-site training for clients including TATA, Mahindra and Volkswagen.
- Built PoCs, optimized cycle times and produced comprehensive handover documentation.

Grindmaster Machines — *R&D Engineer*

Aug 2018 – Oct 2020

- Machine Vision System Integration for Rockman Industries (Tirupati): integrated Cognex A5000 3D vision with ABB robot for bin-picking; implemented TCP/IP sensor messaging and ML classification.
- Wheel Rim Deburring (Endurance Pune): installed robotic deburring cells with KUKA and integrated advanced instrumentation.
- Cylinder Head Robotic Machining (Renault Nissan Chennai): developed gate cutting solution; integrated Fanuc/ABB robots, lasers and PLCs.
- IoT & Polishing (Alicon): developed IIoT polishing/deburring with Beckhoff IPC and ML-based process monitoring; implemented predictive spindle vibration detection.
- COVID Ventilator: developed low-cost ventilator (embedded control, Ambu bag actuation); deployed in hospitals and covered by media.
- Tech: ABB, KUKA, FANUC, Cognex, Mitsubishi/Siemens/Allen Bradley PLCs, Beckhoff, Profinet/EtherCAT/DeviceNet, sensors, OpenCV, TensorFlow/PyTorch.

Education

M.Tech in Nanotechnology, IIT Roorkee

2021 – 2023

Thesis: Advanced Nanomaterials for Medical Robotics Applications

- Research focused on developing nanomaterials for enhanced sensor capabilities in surgical robots
- Published paper on "Nanocomposite Materials for Improved Haptic Feedback in Robotic Surgery"

B.Tech in Electronics & Telecommunication, MIT Aurangabad

2015 – 2018

Final Year Project: Nanolite Advance Control System (Grindmaster)

- Worked on GUI development, Raspberry Pi, Embedded C, ABB integration, IoT, Computer Vision.

Polytechnic in Electronics and Telecommunication, GPA Aurangabad

2013 – 2015

Key Projects

- **Wagon Tippler Automation (Tata Steel):** First robotic wagon tippler unloading solution in India — robot sequencing, safety interlocks, PLC and HMI integration.
- **Cylinder Head Robotic Machining (Renault Nissan):** Laser-based gate cutting system; replaced legacy machine and implemented robust QA and marking processes.
- **Low-cost Ventilator (COVID):** Embedded control, alarm logic, mechanics; successful deployment and media coverage.

Activities & Innovation

- Director — Technology for India Foundation (TFI) — STEM outreach.
- Robocon 2017 — semi-autonomous Frisbee robot (team lead).
- Patent research & drafting — robotics and AI (titles available on request).

References

Mr. Sameer Kelkar – CEO, Grindmaster Machines Pvt. Ltd.

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Dr. Indrajit Desai – PhD, IIT Bombay

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